Variational AutoEncoder

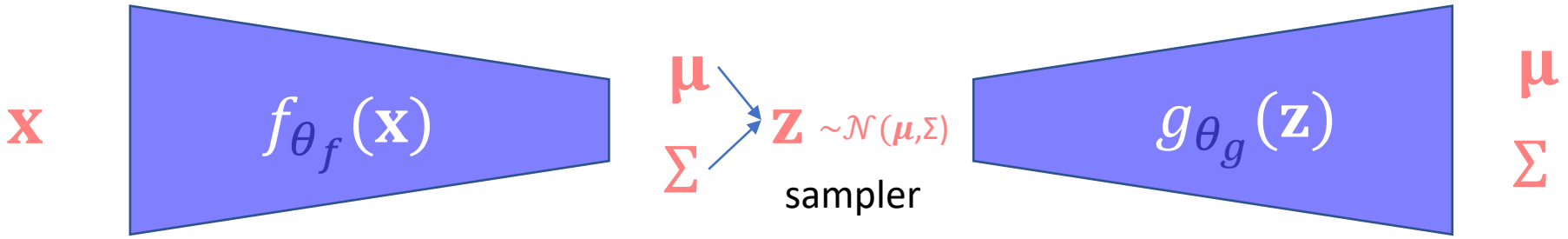
Recap

- Generative modelling goals
  - Generate realistic samples
  - Assign likelihood to samples (Density Estimation)
  - $P(\mathbf{z}|\mathbf{x})$: (Compressed) Representation Learning
    - Understanding underlying generative factors (e.g. similar to what PCA or matrix factorization do)
    - Data compression
    - Semi-supervised learning
    - ....
- Training objective for VAE

$$\max_{\theta_f,\theta_g} \sum_{i=1}^{n} \left(\frac{1}{s} \sum_{k=1}^{s} \left(\log \mathcal{N}\left(\mathbf{x}_i; \{\boldsymbol{\mu}, \Sigma\} = g\left(\widehat{\mathbf{z}_i^k}\right)\right)\right) - D_{KL}\left(\mathcal{N}(\{\boldsymbol{\mu}, \Sigma\} = f(\mathbf{x_i})) \,||\mathcal{N}(\boldsymbol{\mu} = \mathbf{0}, \Sigma = \mathbb{I})\right)\right)$$

$$\mathbf{z} \sim Q_{f(\mathbf{x})}(\mathbf{z}|\mathbf{x})$$

$$\mathbf{x} \sim P_{g(\mathbf{z})}(\mathbf{x}|\mathbf{z})$$



Back to the simple idea!

- How do we train VAEs?
- How do we generate sample from Trained VAEs?
- How do we do density estimation?
- How do we code samples?

VAE - Training

VAE training implementation

$$\max_{\theta_f,\theta_g} \sum_{i=1}^{n} \left(\frac{1}{s} \sum_{k=1}^{s} \left(\log \mathcal{N}\left(\mathbf{x}_i; \{\boldsymbol{\mu}, \Sigma\} = g\left(\widehat{\mathbf{z}_i^k} \right) \right) \right) - D_{KL}\left(\mathbf{z}; \mathcal{N}(\{\boldsymbol{\mu}, \Sigma\} = f(\mathbf{x}_i)) \,||\, \mathcal{N}(\mathbf{z}; \boldsymbol{\mu} = \mathbf{0}, \Sigma = \mathbb{I}) \right) \right)$$

$$\left\{ \widehat{\mathbf{z}_i^k} \right\}_{k:1..s} \sim \mathcal{N}(\mathbf{z}; \{\boldsymbol{\mu}, \Sigma\} = f(\mathbf{x}_i))$$

- Common case in practice
  - $P(\mathbf{z}) = \mathcal{N}(\mathbf{z}; \boldsymbol{\mu} = \mathbf{0}, \Sigma = \mathbb{I})$
  - $P(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\mathbf{x}; \{\boldsymbol{\mu}_\mathbf{x} = g(\mathbf{z}), \mathbb{I}\}$
  - $Q(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\mathbf{z}; \{\boldsymbol{\mu}_\mathbf{z}, \boldsymbol{\sigma}_\mathbf{z}\mathbb{I}\} = f(\mathbf{x}))$
  - Single-sample Monte Carlo estimation ($s = 1$)

$$\min_{\theta_f,\theta_g} \mathcal{L}(\theta_f, \theta_g, X) \triangleq \sum_{i=1}^{n} \sum_{j=1}^{l} \left(\frac{\left(\mu_{\mathbf{x},j} - x_{i,j}\right)^2}{2} - \log \sigma_{\mathbf{z},j} + \frac{\sigma_{\mathbf{z},j}^2 + \mu_{\mathbf{z},j}^2}{2} \right)$$

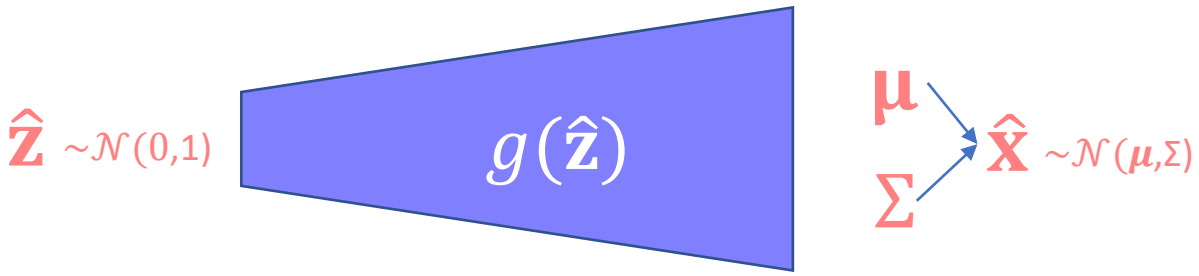
we constructed VAE by following the 3rd goal

1st Goal: How to sample from $P(\mathbf{x})$ using the trained VAE?

VAE – Generative Inference

VAE Generative modeling: steps

- Take the decoder g
- Sample $\hat{\mathbf{z}}$ from the prior *e.g.*, $\mathcal{N}(\mathbf{0}, \mathbb{I})$
- Then sample $\hat{\mathbf{x}}$ from $P(\mathbf{x}|\hat{\mathbf{z}})$ *e.g.*, using mean and variance from $g(\hat{\mathbf{z}})$



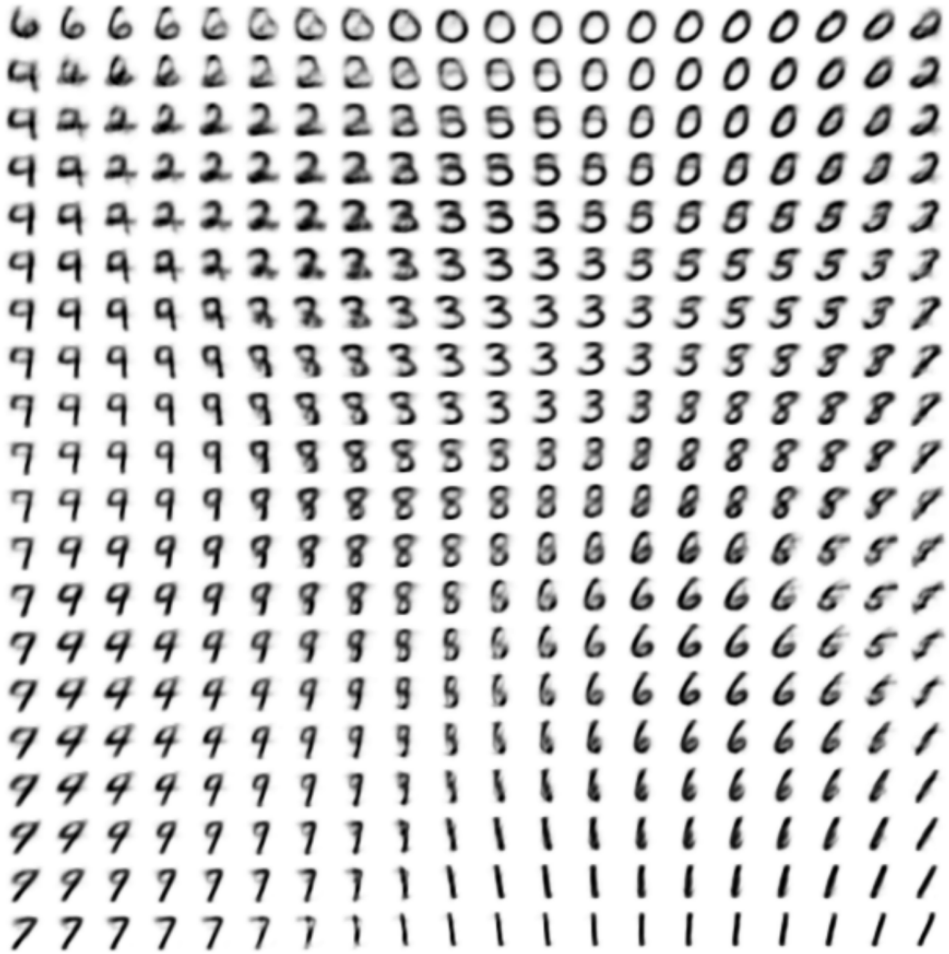
VAE – Generative Inference

VAE Generative modeling: useful tricks

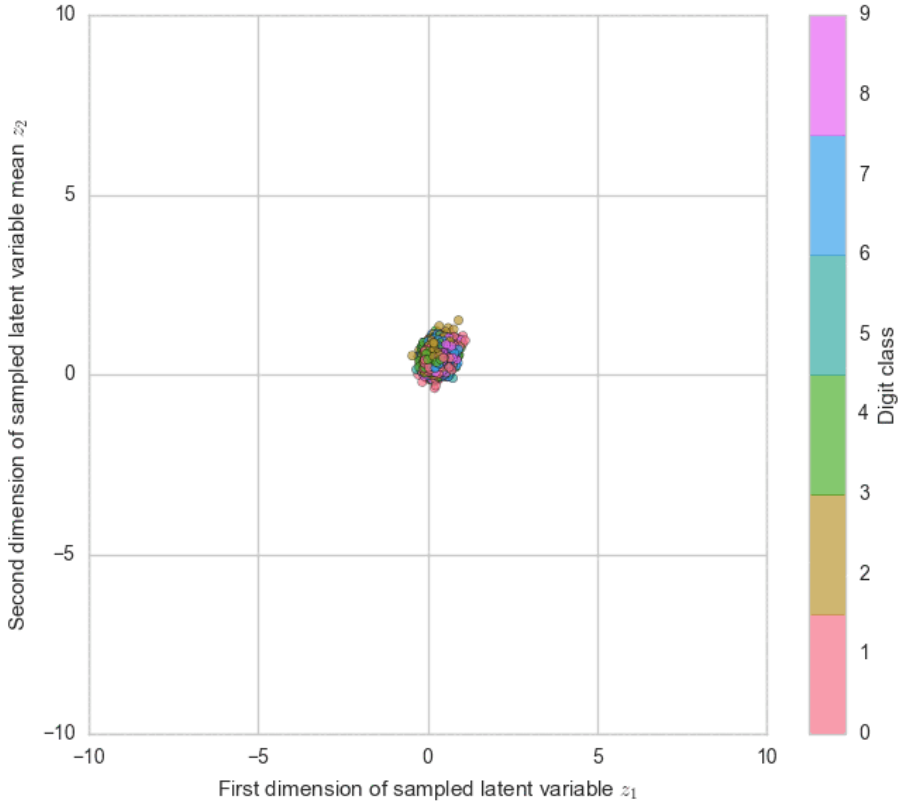
- Variances
  - have the network predict the log of variance instead of the variance itself
  - and/or have the network to have a lower bound for the variance epsilon (never let it to go to zero)
  - fix the variance of the decoder (to 1)
  - learn one standard deviation for all pixels (quite effective in practice, project in WASP from Viktor Nilsson)
    - “Simple and Effective VAE Training with Calibrated Decoders.”, ICML 2021
- At generation time, only use the mean to avoid blurriness
- Use KL Warm-up
  - [Sonderby et al., “Ladder Variational Autoencoders”, NIPS 2016]

VAE – Generative Inference

VAE Generative modeling: results

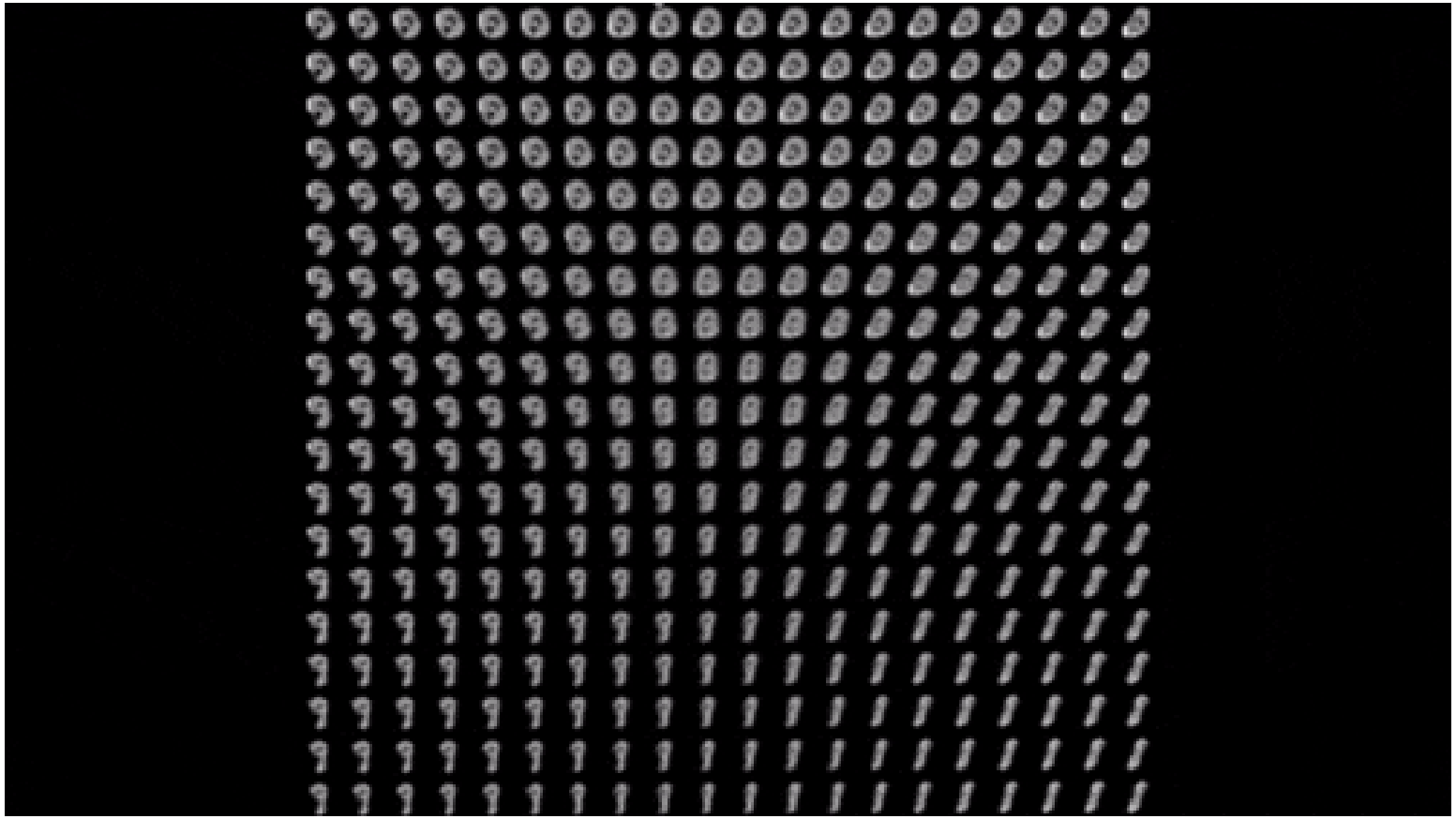


VAE Generative modeling: results



https://jaan.io/what-is-variational-autoencoder-vae-tutorial/

VAE Generative modeling: results



Reflection/Discussion point:

- Remember the VAE gives us an encoder P(z|x) and a decoder P(x|z).
- But, can VAEs be used as a density estimator as well($P(\mathbf{x})$)?



5 minutes!

we constructed VAE by following the 3rd goal

2nd Goal: How to obtain density $P(\mathbf{x})$ for a given input $\mathbf{x}$?

This is useful for OoD detection for instance!

VAE – Density Estimation

VAE Density Estimation: evidence lower bound

You can use the ELBO!

$$\log P(\mathbf{x}) = D_{KL}(Q_{\boldsymbol{\omega}}(\mathbf{z})||P(\mathbf{z}|\mathbf{x})) + \sum Q_{\boldsymbol{\omega}}(\mathbf{z})\log\frac{P(\mathbf{x},\mathbf{z})}{Q_{\boldsymbol{\omega}}(\mathbf{z})}$$

VAE – Density Estimation

VAE Density Estimation: standard procedure

- $P(\mathbf{x}) = \int P(\mathbf{x}|\mathbf{z})P(\mathbf{z})d\mathbf{z}$
- Monte Carlo Sampling
  - Sample $\{\mathbf{z}_l\}_{l:1\ldots\mathrm{L}} \sim P(\mathbf{z})$
  - $P(\mathbf{x}) \approx \frac{1}{\mathrm{L}}\sum P(\mathbf{x}|\mathbf{z}_l)$

very large variance with low L

VAE – Density Estimation

VAE Density Estimation: importance sampling

- Sample $\{\mathbf{z}_l\}_{l:1\ldots\mathrm{L}} \sim Q(\mathbf{z}|\mathbf{x})$
- $P(\mathbf{x}) \approx \frac{1}{\mathrm{L}} \sum \frac{P(\mathbf{x},\mathbf{z}_l)}{Q(\mathbf{z}_l|\mathbf{x})}$

better variance, can be good estimate for low-dimensional z

VAE – Density Estimation

VAE Density Estimation: importance sampling

- Sample $\{\mathbf{z}_l\}_{l:1\ldots\mathrm{L}} \sim Q(\mathbf{z}|\mathbf{x})$
- $P(\mathbf{x}) \approx \frac{1}{\mathrm{L}} \sum \frac{P(\mathbf{x},\mathbf{z}_l)}{Q(\mathbf{z}_l|\mathbf{x})}$

Remember VI ELBO was $\sum Q(\mathbf{z}) log \frac{P(\mathbf{x},\mathbf{z})}{Q(\mathbf{z})}$

[Burda, Grosse, Salakhutdinov, "Importance Weighted Autoencoders", ICLR 2016]

3rd Goal: Well, we know that, and VAEs are good for learning a latent representation